Lab 3

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1.1

3.

$$Y_i = 1.478X_i + \epsilon_i$$

4.

Forcing the line to go through the origin does **not** seem appropriate for this data set.

5.

The pattern seems relatively flat, maybe a slight positive relationship as theres a larger clustering around the top on the righthand side than the left.

6.

The residuals do not sum to zero, they sum to 66.27

Part B

2.

```
standard_reg <- lm(Rent.Rate ~ Op.Expense, data = data)
X <- model.matrix(standard_reg)

solve(t(X) %*% X)
```

```
            (Intercept)   Op.Expense
(Intercept)  0.18817287 -0.018148689
Op.Expense  -0.01814869  0.001873288
```

4.

```r
Y <- data$Rent.Rate
b <- solve(t(X) %*% X) %*% (t(X) %*% Y)
b
```

```
                  [,1]
(Intercept) 12.4702588
Op.Expense   0.2754531
```

The coefficients in $\vec{b}$ are the same as the output of summary.

5.

```r
y_hat <- X %*% b
y_hat |> head()
```

```
      [,1]
1 13.85303
2 14.72622
3 13.29662
4 15.41761
5 14.94107
6 15.07329
```

These values are the same as the fitted values from the regression.

6.

```r
(Y - y_hat) |> head()
```

```
        [,1]
1 -0.3530332
2 -2.7262194
3 -2.7966180
4 -0.4176066
5 -0.9410728
6 -4.5732903
```

These match the residuals of the regression.

7.

```
sqrt(1.575^2 * solve(t(X) %*% X))
```

```
Warning in sqrt(1.575^2 * solve(t(X) %*% X)): NaNs produced
```

```
            (Intercept) Op.Expense
(Intercept)   0.6832176        NaN
Op.Expense          NaN 0.06816835
```

The diagonals are the same as the std. error from the output of summary.

Part C

```
length <- c(63, 63, 63, 63, 63, 63, 63, 63, 63)
weight<- c(136, 198, 194, 140, 93, 172, 116, 174, 145)
snakes <- data.frame(Length = length, Weight = weight)

my.reg <- lm(Weight ~ Length, data = snakes)
```

There's only 1 unique value for length, so that ruins everything here.